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Best of luck!

PAST PAPER

Question No : 1 of 26 Marks: 1 (Budgeted Time 1 Min)

The form of the exact solution to

$$2\frac{dy}{dx} + 3y = e^{-x}, y(0) = 5$$

is

Answer (Please select your correct option)

- $Ae^{-1.5x} + Bxe^{-x}$
- $Ae^{1.5x} + Be^{-x}$
- $Ae^{1.5x} + Bxe^{-x}$
- $Ae^{-1.5x} + Be^{-x}$ ⇦ correct answer solve by hadi

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Question No : 2 of 26 **Marks: 1 (Budgeted Time 1 Min)**

The Period of oscillator in the solution X=50Sin (20t+8.5) is

Answer (Please select your correct option)

- 0.17643
- 0.32045
- 0.31400
- 0.58000

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Question No : 2 of 26 Marks: 1 (Budgeted Time 1 Min)

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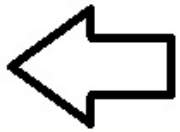
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Question No : 3 of 26 Marks: 1 (Budgeted Time 1 Min)

A problem of the differential equation associated with a single value of the unknown function at a given point in the domain of the solution is known as ----------

Answer (Please select your correct option)

- Boundary Value Problem
- Initial Value Problem
- None of these



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Question No : 4 of 26 Marks: 1 (Budgeted Time 1 Min)

The equation $\frac{dy}{dx} = \frac{x(x+1)}{y(y-1)}$ is a/an ----------

Answer (Please select your correct option)

- partial differential equation.
- ordinary differential equation.
- polynomial equation.
- transcendental equation.



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Question No : 5 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If $f(x,y)=0$, then $\frac{dy}{dx}=....$

Answer (Please select your correct option)

- $-\frac{f_x}{f_y}$
- $-\frac{f_y}{f_x}$
- $\frac{f_x}{f_y}$
- $\frac{f_y}{f_x}$

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Question No : 6 of 26 Marks: 1 (Budgeted Time 1 Min)

For $y^2(1-x^2)-\cos^2 x = c$ *with* $y(0) = 3$, *the value of* c *is* ------.

Answer (Please select your correct option)

- 9
- 8
- 10
- 5



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Question No : 7 of 26 Marks: 1 (Budgeted Time 1 Min)

The differential equation $(3x^2y + 2)\,dx + (x^3 + y)\,dy = 0$ is -------.

Answer (Please select your correct option)

- Exact



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- Non-exact
- Separable
- Homogenous

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Question No : 8 of 26 Marks: 1 (Budgeted Time 1 Min)

For a 1st order linear differential equation $\frac{dy}{dt} - \frac{1}{t}y = 2t$; the continuous function, $q(t)$, is.....

Answer (Please select your correct option)

- $2t$
- $-\frac{1}{t}$
- $\frac{1}{t}$
- t

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Question No : 9 of 26 **Marks: 1 (Budgeted Time 1 Min)**

For a Ist order linear differential equation $\frac{dy}{dx}(x+2y^2)=y$; The continuous function, $q(y)$, is---.

Answer (Please select your correct option)

- x^2
- $2y$ ⇦ correct answer solve by hadi
- $2y^2$
- $-2y$

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Question No : 10 of 26 Marks: 1 (Budgeted Time 1 Min)

A differential equation $\frac{dy}{dx} + p(x)y = q(x)y^n$ *for* $n \neq 0,1$ is called a/an ----.

Answer (Please select your correct option)

- Non-Exact differential equation
- Linear differential equation
- Bernoulli equation ⇦ correct answer solve by hadi
- Exact differential equation

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Question No : 11 of 26 Marks: 1 (Budgeted Time 1 Min)

The constant solutions of the logistic equation $\frac{dP}{dt} = P(a - bP)$ are----.

Answer (Please select your correct option)

- P=0, P= b/a
- P=0, P=a/b
- P=a/b, P=b/a
- P=0,P=0



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Question No : 12 of 26 Marks: 1 (Budgeted Time 1 Min)

The logistic equation $\frac{dP}{dt} = P(a - bP)$ is a/an------.

Answer (Please select your correct option)

- Linear differential equation
- Non-linear differential equation  correct answer solve by hadi
- Homogenous differential equation
- Exact differential equation

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Question No : 13 of 26 Marks: 1 (Budgeted Time 1 Min)

Wronskian of $Sin\ x, Cos\ x$ is----.

Answer (Please select your correct option)

- Sinx
- 1
- -1
- Cosx



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Question No : 14 of 26 **Marks: 1 (Budgeted Time 1 Min)**

What is annihilator operator of the function $\frac{dy}{dx} + 4y = e^{2x}$?

Answer (Please select your correct option)

- $(D+4)(D-2)$
- $(D-2)$
- $(D+2)$
- $D(D^2-2)$

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Question No : 15 of 26 Marks: 1 (Budgeted Time 1 Min)

What is the annihilator operator of the function $g(x) = \cos 2x$?

Answer (Please select your correct option)

- $(D-4)$
- $(D+4)$
- (D^2+4)
- (D^3-4)

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Question No : 16 of 26 Marks: 1 (Budgeted Time 1 Min)

The differential equation of the orthogonal trajectory to the family of curves $y = cx$ is-----.

Answer (Please select your correct option)

- $\frac{dy}{dx} = -\frac{x}{y}$
- $\frac{dy}{dx} = \frac{x}{y}$
- $\frac{dy}{dx} = -\frac{y}{x}$
- $\frac{dy}{dx} = \frac{y}{x}$ ⇦

correct answer s
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Question No : 17 of 26 Marks: 1 (Budgeted Time 1 Min)

The orthogonal trajectory to the family of curves $3x + 4y = c$ is------.

Answer (Please select your correct option)

- $y = 2x$
- $y - x = c$
- $4x - 3y = c$
- $4x + 3y = c$



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Question No : 18 of 26 Marks: 1 (Budgeted Time 1 Min)

In the population growth model, the solution of the differential equation is given by .....

Answer (Please select your correct option)

- $A(t) = ke^{kt}$
- $A(t) = A_0 e^{kt}$ 
- $A(t) = A_0 / 2$
- $A(t) = ke^{A_0 t}$

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Question No : 19 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If $y_2 = x^3$ is the second solution of $x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} - 3y = 0$, then which of the following is the most accurate option?

Answer (Please select your correct option)

- The first solution say y_1 must be non-trivial on the indicated interval.
- The first solution say y_1 can be trivial on the indicated interval.
- The first solution say y_1 must be discontinuous on the indicated interval.
- The first solution say y_1 may or may not be continuous on the indicated interval.

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Question No : 20 of 26 Marks: 1 (Budgeted Time 1 Min)

If the complementary solution of $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 3\sin x$ is $c_1e^{-x} + c_2e^{2x}$, then which of the following is the most accurate option for it?

Answer (Please select your correct option)

- The general form of the particular solution will be $(A\sin x + B\cos x)e^x$.
- The general form of the particular solution will be $Ax\sin x + Bx\cos x$.
- The general form of the particular solution will be $(Ax\sin x + Bx\cos x)e^{-x}$.
- The general form of the particular solution will be $A\sin x + B\cos x$.



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Question No : 21 of 26 **Marks: 2 (Budgeted Time 4 Min)**

If a mass weighing $20lb$ stretches a spring by $\frac{1}{2}\ ft$ then find k using the Hooke's Law.

Answer (Please click here to Add Answer)

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Question No : 21 of 26 Marks: 2 (Budgeted Time 4 Min)

If a mass weighing 20*lb* stretches a spring by ½ *ft* then find *k* using the Hooke's Law.

Answer (Please click here to Add Answer)

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Question No : 22 of 26 **Marks: 2 (Budgeted Time 4 Min)**

Show that $y = x^2$ is a solution of the following differential equation:

$$x^2 \frac{d^2y}{dx^2} - 3x \frac{dy}{dx} + 4y = 0$$

Answer (Please click here to Add Answer)

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Question No : 22 of 26 Marks: 2 (Budgeted Time 4 Min)

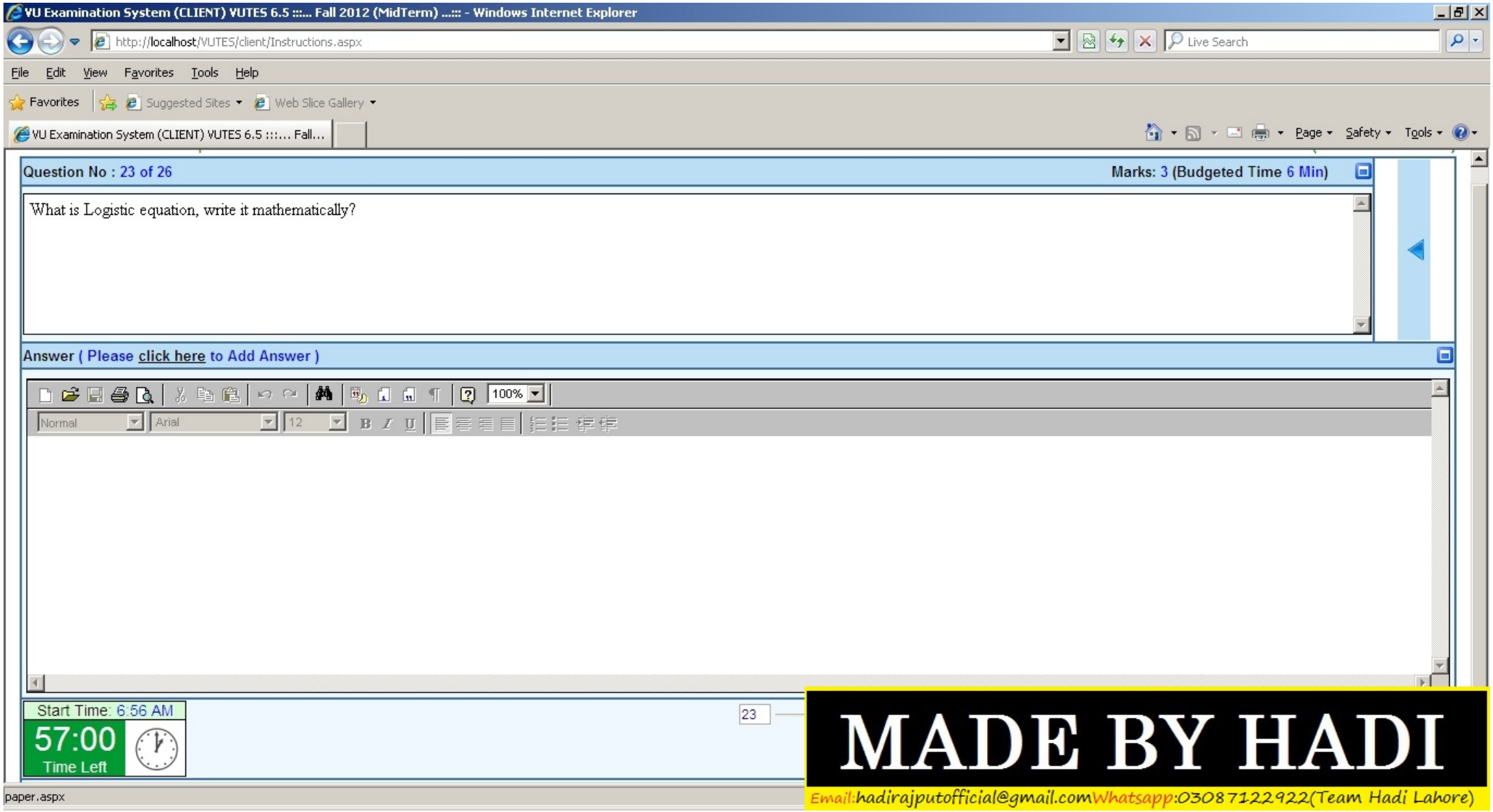
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$$x^2 \frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 4y = 0$$

Answer (Please click here to Add Answer)

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Question No : 23 of 26 Marks: 3 (Budgeted Time 6 Min)

What is Logistic equation, write it mathematically?

Answer (Please click here to Add Answer)



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Question No : 24 of 26 **Marks: 3 (Budgeted Time 6 Min)**

Write the following differential equation in the form $L(y) = g(x)$, where L is a linear differential operator with constant coefficients.

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 5y = 2\sin y$$

Answer (Please click here to Add Answer)

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Question No : 24 of 26 Marks: 3 (Budgeted Time 6 Min)

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$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 5y = 2\sin y$$

Answer (Please click here to Add Answer)

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Question No : 25 of 26 **Marks: 5 (Budgeted Time 10 Min)**

Find the wronskian of the Differential equation $y'''-2y''-21y'-18y=3+4e^{-t}$ using variation of parameter and the root of the auxiliary equation is $m_1=-3, m_2=-1, m_3=6$?

Answer (Please click here to Add Answer)

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Question No : 25 of 26 Marks: 5 (Budgeted Time 10 Min)

Find the wronskian of the Differential equation $y'''-2y''-21y'-18y=3+4e^{-t}$ using variation of parameter and the root of the auxiliary equation is $m_1=-3, m_2=-1, m_3=6$?

Answer (Please click here to Add Answer)



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Question No : 26 of 26 **Marks: 5 (Budgeted Time 10 Min)**

Solve the boundary value problem $\frac{d^2y}{dx^2} + 25y = 0$ subject to the boundary conditions $y(0) = 0$, $y\left(\frac{\pi}{10}\right) = 0$

Answer (Please click here to Add Answer)

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Question No : 26 of 26 Marks: 5 (Budgeted Time 10 Min)

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Answer (Please click here to Add Answer)

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